

Set 2: Customer Feedback Analysis

```
install.packages("ggplot2")
install.packages("plotly")
install.packages("wordcloud")
install.packages("RColorBrewer")
library(ggplot2)
library(plotly)
customer_data <- data.frame(
  Customer_ID = c(1, 2, 3, 4, 5),
  Age = c(25, 30, 35, 28, 40),
  Satisfaction_Score = c(4, 5, 3, 4, 5)
)
print(customer_data)
hist(
  customer_data$Age,
  main = "Distribution of Customer Ages",
  xlab = "Age",
  ylab = "Number of Customers",
  breaks = 5
)
satisfaction_count <- table(
  customer_data$Satisfaction_Score
)
labels <- paste(
  names(satisfaction_count),
  "Score"
)
pie(
  satisfaction_count,
  labels = labels,
  main = "Distribution of Customer Satisfaction Scores"
```

```
)
```

```
customer_data$Age_Group <- cut(  
  customer_data$Age,  
  breaks = c(20, 29, 39, 50),  
  labels = c("20-29", "30-39", "40-50")  
)
```

```
# Create table
```

```
age_satisfaction <- table(  
  customer_data$Age_Group,  
  customer_data$Satisfaction_Score  
)
```

```
barplot(  
  age_satisfaction,  
  beside = FALSE,  
  main = "Satisfaction Scores by Age Group",  
  xlab = "Age Group",  
  ylab = "Number of Customers",  
  legend.text = TRUE  
)
```

```
feedback <- c(  
  "excellent service and very helpful staff",  
  "very satisfied with the excellent service",  
  "good service and friendly staff",  
  "average experience but good support",  
  "excellent product and fast service"  
)
```

```
library(tm)
```

```
library(wordcloud)
library(RColorBrewer)
corpus <- Corpus(
  VectorSource(feedback)
)
corpus <- tm_map(
  corpus,
  content_transformer(tolower)
)
corpus <- tm_map(
  corpus,
  removePunctuation
)
corpus <- tm_map(
  corpus,
  removeNumbers
)

corpus <- tm_map(
  corpus,
  removeWords,
  stopwords("english")
)

tdm <- TermDocumentMatrix(corpus)
m <- as.matrix(tdm)
word_frequency <- sort(
  rowSums(m),
  decreasing = TRUE
)
wordcloud(
```

```

words = names(word_frequency),

freq = word_frequency,

min.freq = 1,

max.words = 50,

random.order = FALSE,

scale = c(4, 0.5)

)

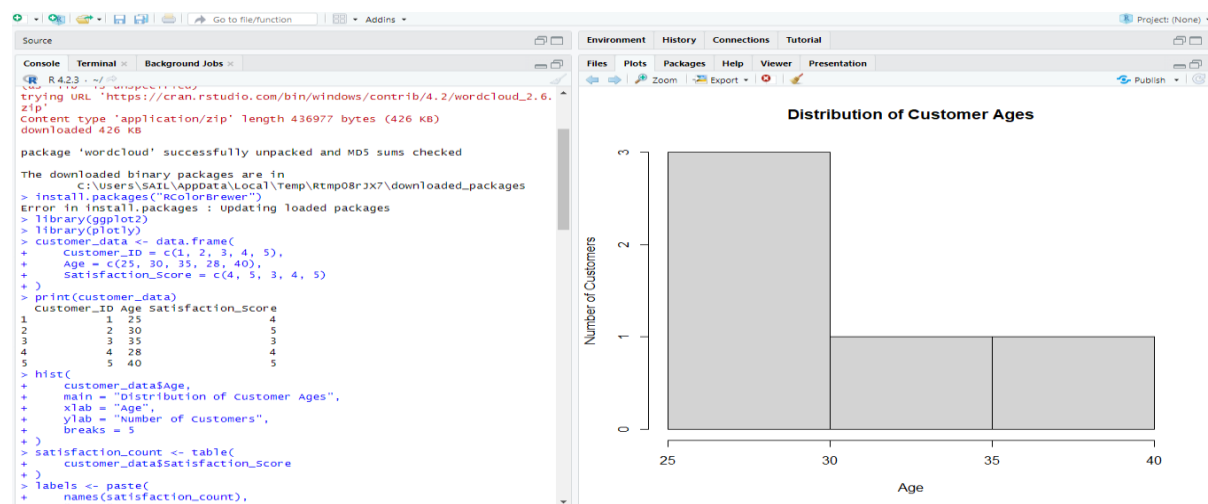
title(

  "Customer Feedback Sentiment Word Cloud"

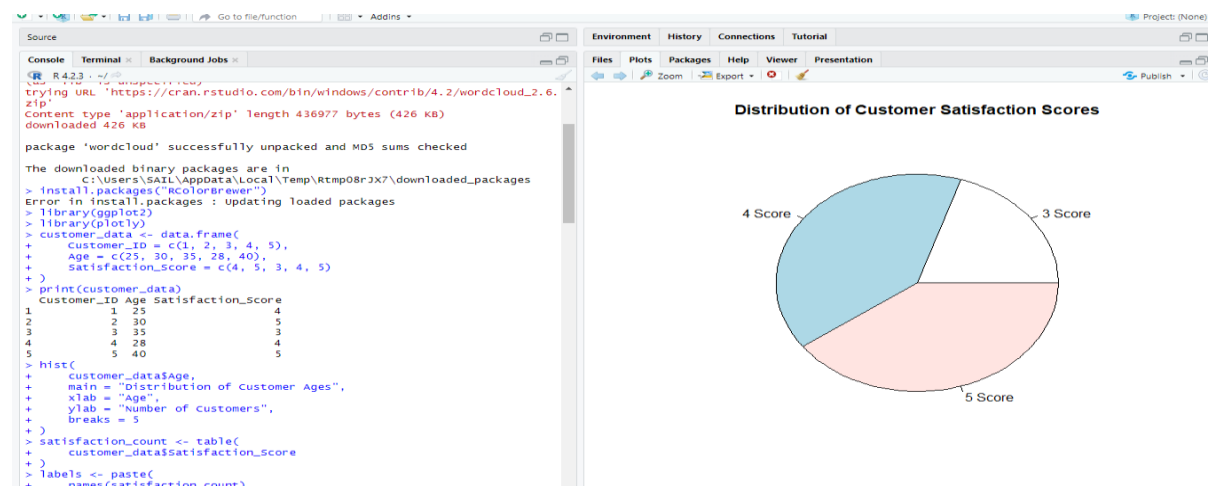
)

```

1. Create a histogram to represent the distribution of customer ages. Label the axes and title the chart.



2. Generate a pie chart to display the overall distribution of customer satisfaction scores. Include labels.



3. Build a stacked bar chart to visualize the distribution of customer satisfaction scores by age group.



4. Develop a word cloud from open-ended customer feedback to identify prevalent customer sentiments.

